Lesson for the course “Human-Computer Interaction”

# Design and prototyping

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# Conceptual and concrete design

- There are two aspects to design:

1. **The conceptual part:** focuses on the idea of a product
   - It involves developing a conceptual model that captures what the product will do and how it will behave
2. **The concrete aspect:** focuses on the details of the design
   - E.g., menu types, haptic feedback, physical widgets, and graphics

- The two are intertwined:
  - Design issues will require some consideration in order to prototype ideas
  - Prototyping ideas will lead to an evolution of the concept

# Conceptual design

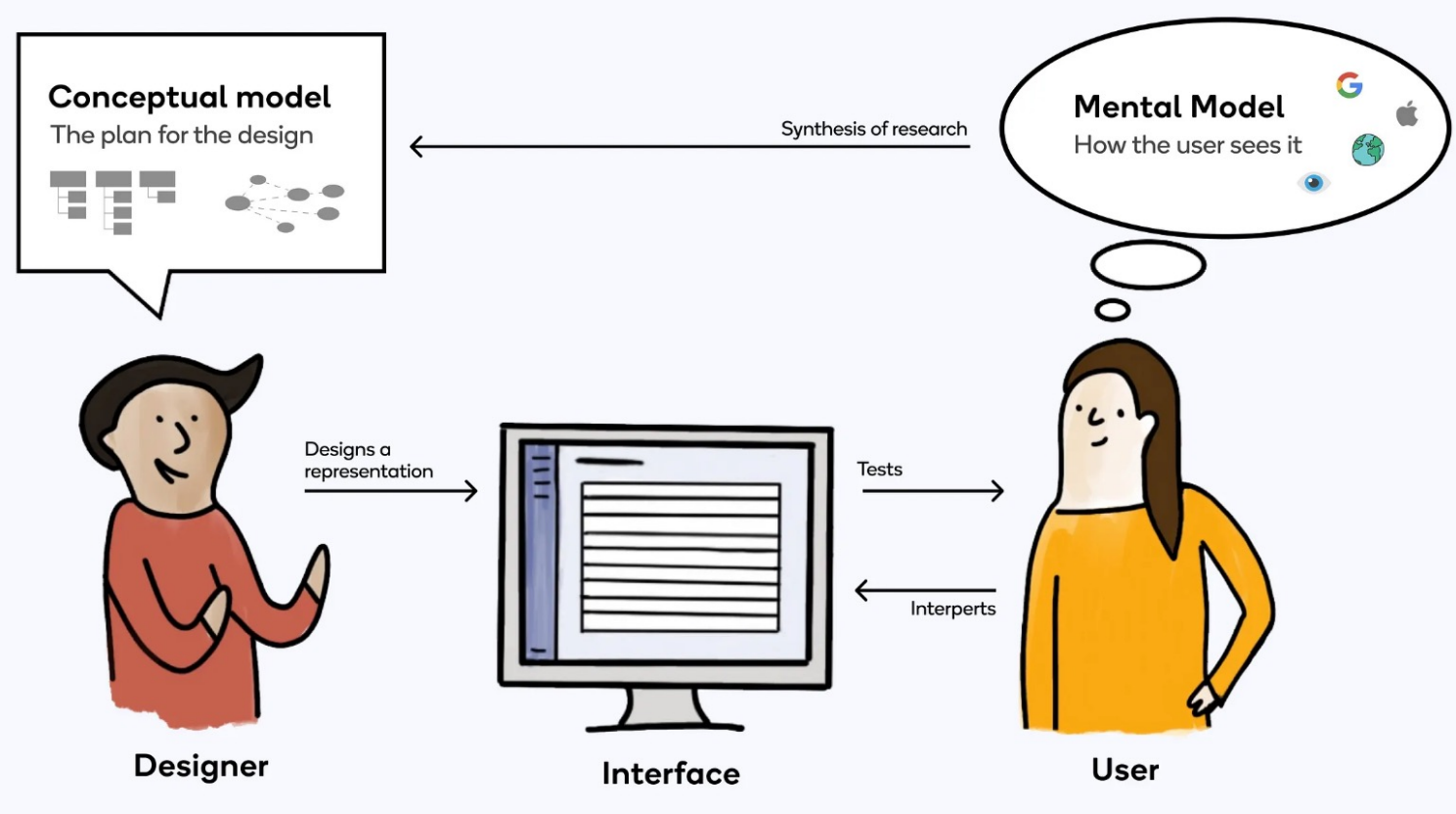
- Is concerned with developing a **conceptual model** of the user-product interaction
- A conceptual model is:
  - An outline of what people can do with a product
  - What concepts are needed to understand and interact with it
- To develop the conceptual model you need to:
  - Understand the problem space and the current functional requirements
  - Immerse yourself in the data about the users and their goals, and try to empathize with them (e.g., via field studies)

# Conceptual design

- As a result of those activities you generate various ideas about the interaction
- Then you can use **brainstorming techniques** to explore such ideas with other members of the team in order to build a picture of the users and their goals
  - **Gradually**, a picture of the desired users' experience will emerge and become more concrete
- You can also develop a range of **scenarios** that can help to think through the consequences of the different ideas, i.e., to explore different **alternatives**
  - Assess the positive and negative consequences of a particular proposed design solution

# Conceptual design

- The best conceptual models are often those that appear **obvious and simple** and **the operations they support are intuitive to use**



# Developing an initial conceptual model

- The core components of the conceptual model are:
  - The **metaphors**
  - The **concepts** to which users are exposed
  - The **relationship** between those concepts
  - The **mappings** between the concepts and user experience being supported
- **How** the various metaphors, concepts and their relationships are organized **determines the UX**
- The concepts involved in a task and their relationships derive from the product's requirements (e.g., through scenarios and use cases)
- Suitable metaphors will be informed by immersion in the data and attempting to understand the users' perspectives

# Developing an initial conceptual model

- Various approaches exist that help to produce an initial conceptual model
- We consider the following:
  - How to choose **interface metaphors** that will help users understand the product?
  - Which **interaction type(s)** would best support the users' activities?
  - Do different interface types **suggest alternative design** insights or options?
- All of these approaches provide **different ways of thinking** about the product and help generate potential conceptual models

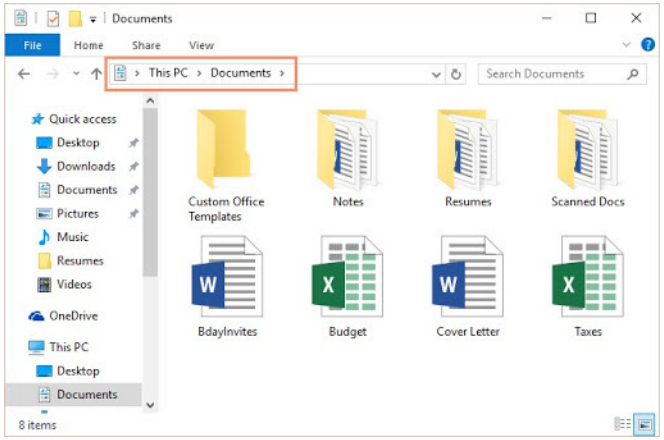
# Metaphors

- **Metaphors** are commonly used to explain something that is unfamiliar or hard to grasp by way of comparison with something that is familiar and easy to grasp
  - E.g., teachers use them to introduce something new to students by comparing the new material with something they already understand
- Metaphors are widely used in IxD to conceptualize abstract, hard-to-imagine, and difficult-to-articulate computer-based concepts and interactions in more concrete and familiar terms

# Interface metaphors

- **Interface metaphors** combine familiar knowledge with new knowledge in a way that will help the user understand the product
- An example of an interface metaphor is the file and folder analogy for the file system of an operating system. Another example is the tree view representation of a file system, as in a file manager.





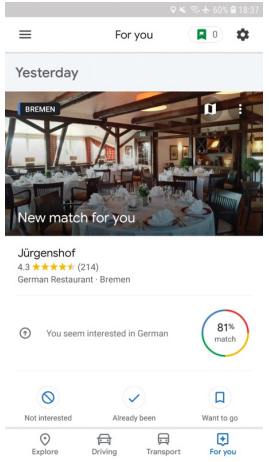
# Examples of interface metaphors

- Metaphors are used in 3 main ways:

1. **As a way of conceptualizing what we are doing**
   - E.g., surfing the web




2. **As a conceptual model instantiated at the interface level**
   - E.g., the card or the desktop metaphors

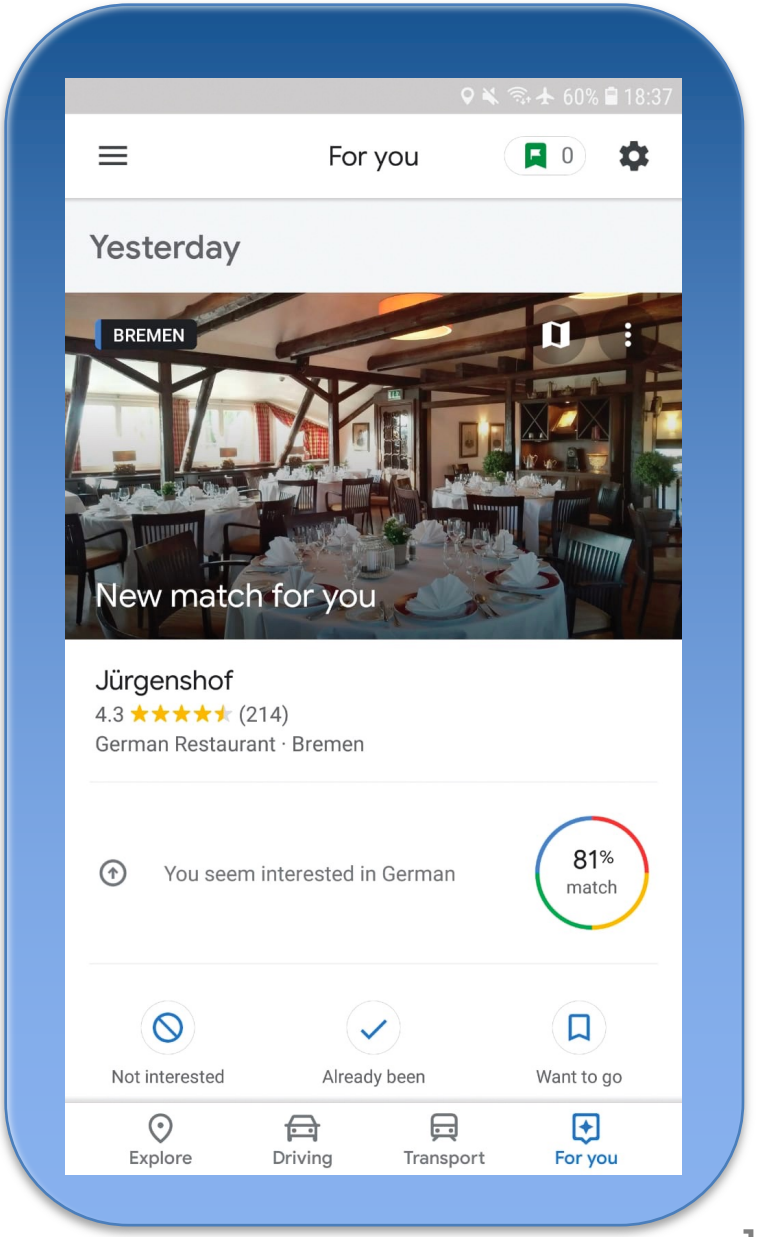



3. **As a way of visualizing an operation**
   - E.g., an icon of a shopping cart into which items are placed that users want to purchase on an online shopping site



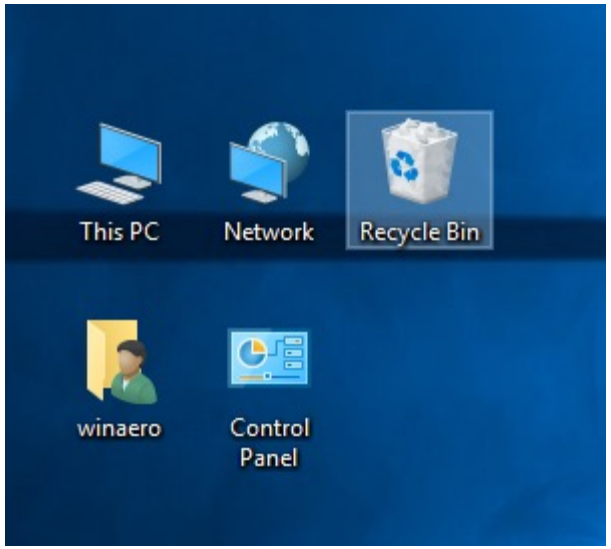
# The card metaphor

- The card is a **very popular** metaphor
- Why?
- Has a familiar form factor
  - E.g., business cards, credit cards, playing cards…
- Can easily be flicked through, sorted, and themed
- Can structure content into meaningful chunks



# Interface metaphors

- **Benefits**
  - Make learning new systems easier
  - Help users understand the underlying conceptual model
  - Can be very innovative and enable the realm of computers and their applications to be made more accessible to a greater diversity of users
- However, may break conventional and cultural rules




- Logically the recycle bin should be placed under the desk
- But users would not have been able to see it because it would have been hidden by the desktop surface, so, it needed to go on the desktop
- Once people understood why the recycle bin icon was on the desktop, they simply **accepted** it being there

# Choosing an interface metaphor

- Choosing suitable metaphors and combining new and familiar concepts requires **a balance between utility and relevance**
- Three steps for choosing a good interface metaphor:

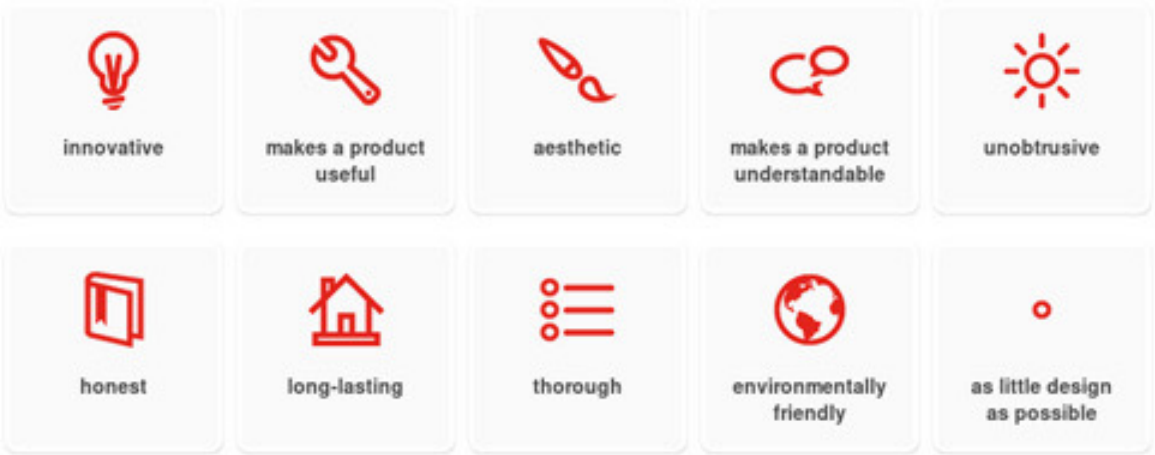
1. **Understand the functionality you want to represent**
   - i.e., functional requirements
2. **Identify potential problem areas related to the functionality**
   - a metaphor can be chosen to support those aspects
3. **Generate metaphors**
   - Search for metaphors in the users' description of the relevant activities, or identify metaphors already used in the application domain

# Evaluating an interface metaphor

- Evaluate the generated metaphors:

1. How much of the metaphor is relevant to the problem?
2. Is it easy to represent?
3. Will the audience understand it?

- **Evaluate with users**



# Tip for the exam

- Try to use interface metaphors and especially try to create new ones
- Example from previous exams: a tree that grows according to the sustainability level achieved during good user's behaviours tracked by an app



# Considering interaction types

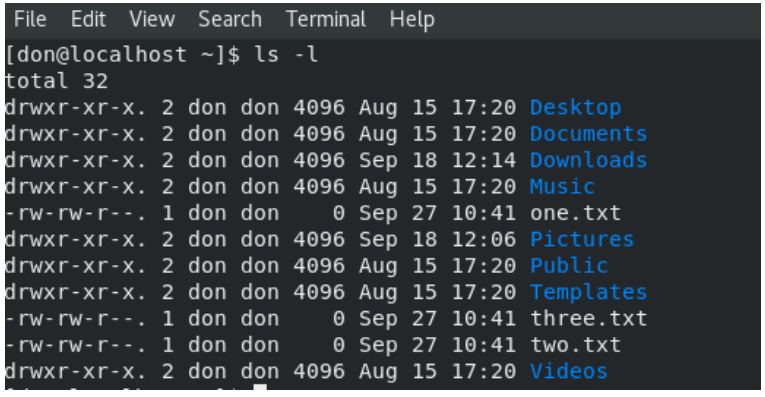
- **Which interaction type best supports users' activities?**

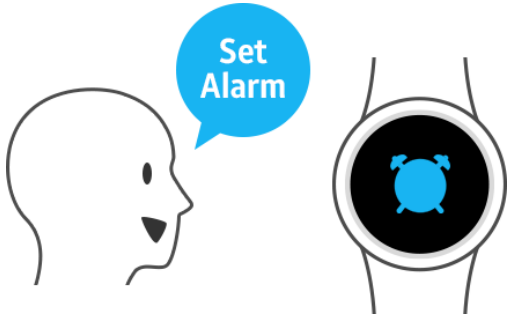
- There are 5 types of interaction:
  1. **Instructing**
  2. **Conversing**
  3. **Manipulating**
  4. **Exploring**
  5. **Responding**

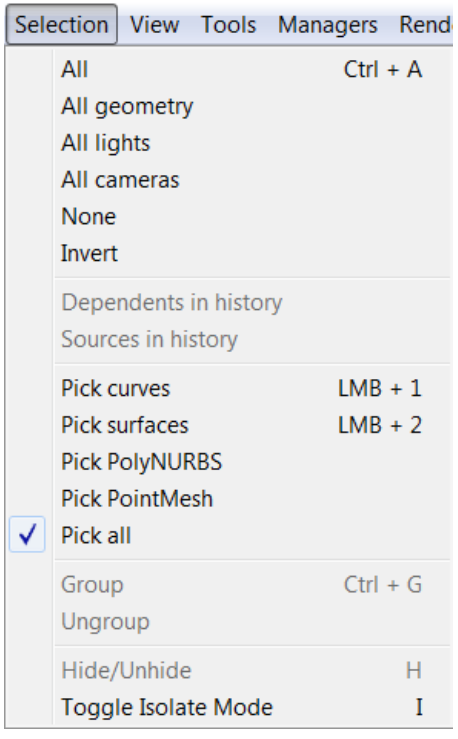
- Most conceptual models will include a **combination** of interaction types (and different parts of the interaction will be associated with different types of interaction)

# Considering interaction types

- **Instructing**: users issue instruction to a system
  - E.g., typing in commands, selecting options from menus in a windows environment or on a multitouch screen, speaking aloud commands, gesturing, pressing buttons, or using a combination of function keys

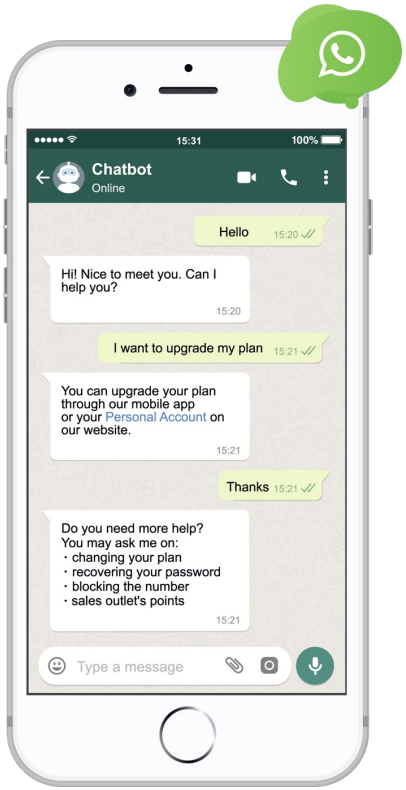


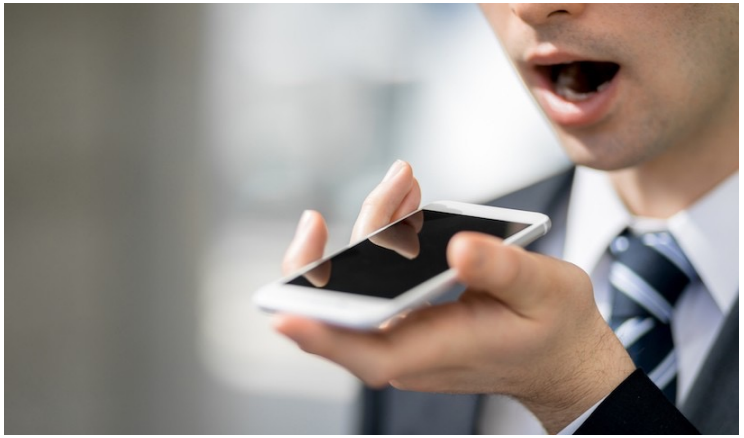



# Considering interaction types

- **Conversing**: users have a dialog with a system
  - E.g., Users can speak via an interface or type in questions to which the system replies via text or speech output



# Considering interaction types

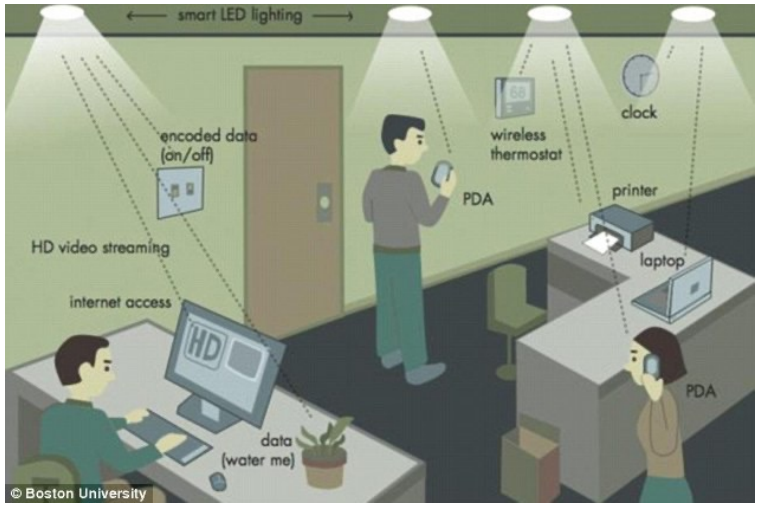
- **Manipulating**: users interact with objects in a virtual or physical space by manipulating them
  - E.g., opening, holding, closing, and placing



# Considering interaction types

- **Exploring**: users move through a virtual environment or a physical space
  - Virtual environments include 3D worlds and augmented and virtual reality systems, which enable users to move around
  - Physical spaces that use sensor-based technologies include smart rooms and ambient environments





# Considering interaction types

- **Responding**: the system initiates the interaction and the user chooses whether to respond
  - E.g., proactive mobile location-based technology can alert people to points of interest (e.g., a restaurant when moving on a street): users can choose to look at the information popping up on their phone or ignore it



# Considering different interface types

- **Do different interface types provide insight?**
  - Graphical user interfaces
  - Tangible user interfaces
  - Augmented or virtual reality
  - Voice-based user interfaces
  - Collaborative user interfaces (designed for more than one person to use)
- Considering different interfaces has both a design and a practical purpose
  - When thinking about the conceptual model for a product, it is important not to be excessively influenced by a predetermined interface type
  - **Different interface types** prompt and support different perspectives on potential user experiences and possible behaviors, hence **prompting alternative design ideas**

# Exercise

- Design different types of discrete or continuous interactions a user can have with a single push button



# Exercise

- Design different types of discrete or continuous interactions a user can have with a 2 push buttons



# Expanding the initial conceptual model

- For prototyping and testing, the ideas generated in the initial conceptual model need some expansion
- **What functions will the product perform?**
  - What will the product do and what will the human do?
- **How are the functions related to each other?**
  - Sequential or parallel?
  - Categorizations (e.g., all actions related to privacy on a smartphone)?
- **What information is needed?**
  - What data is needed to perform the task?
  - How is this data to be transformed by the system?

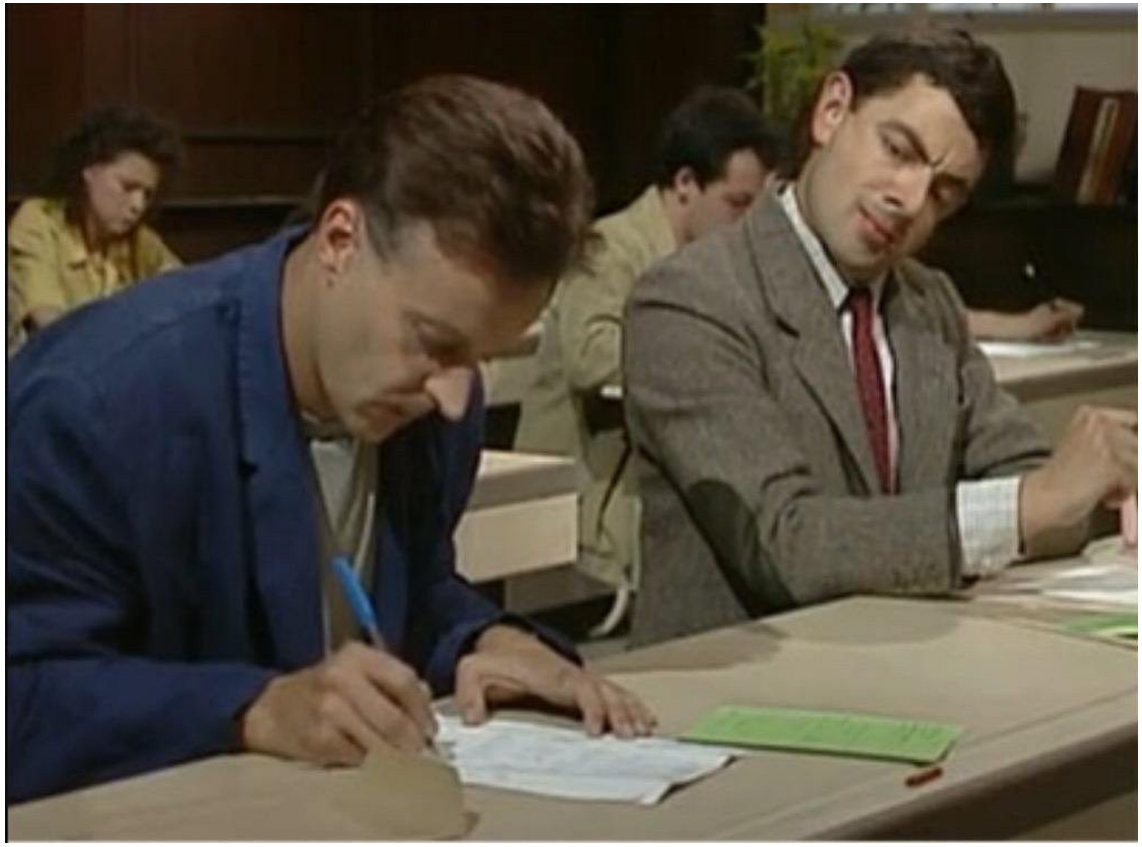
# Concrete design

- Many aspects to design
  - E.g., color, icons, buttons, layout, interaction devices, etc.
- Need to consider issues related to users' characteristics and context
  - **Inclusiveness** (means being fair, open and equal to everyone)
  - **Accessibility** (means the extent to which a product is accessible to as many people as possible)
  - **Cross-cultural design**
    - Language, colors, icons, information architecture
    - Indigenous knowledge and perspectives

# How can one become a good designer?

***Good artists copy, great artists steal***

- Pablo Picasso



# How can one become a good designer?

- **Good designers copy, great designers steal**



Dieter Rams, 1958 Braun T3




Jonathan Ive, 2001, Apple iPod

Listen to the enlightening words of the famous designer Dieter Rams:
http://www.youtube.com/watch?v=-N5aQyCzm4I

# Good practices of a good designer

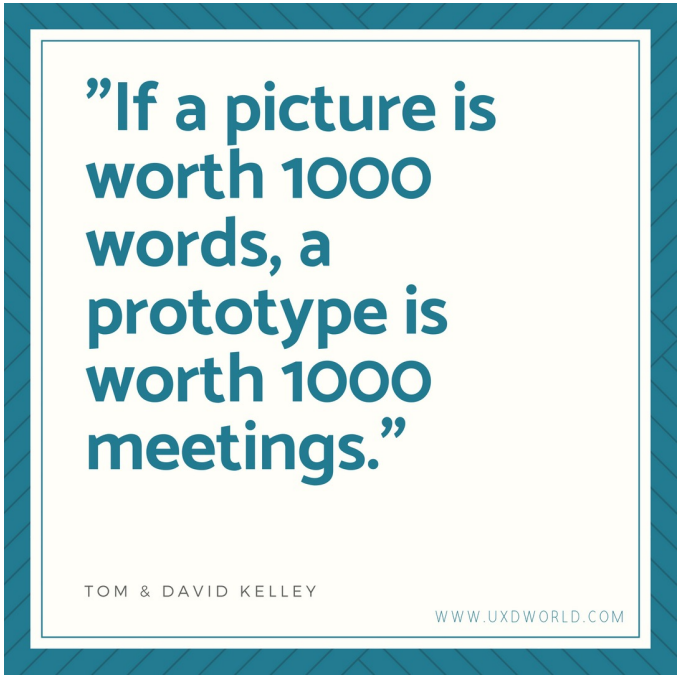
- **Believe in the techniques**
  - When you know how to use them you can violate them
- **Keep a library of good designs**
  - Develop the sensitivity for good design
- **Don't find shortcuts**
  - Start from the users' needs and pain points, explore well the problem space
- **Re-iterate**
  - "Fail early to succeed sooner"
- **Put passion**
  - Into design and communication

# Prototyping

- In the IxD process, solutions or concepts are created, prototyped, tested, and iterated
- The final product emerges **iteratively** through repeated design-evaluation-redesign cycles involving users
  - **Prototypes facilitate this process**
- It is often said that *users can't tell you what they want, but when they see something and get to use it, they soon know what they don't want*

# Prototyping

- Prototyping provides a concrete manifestation of an idea that allows designers to **communicate** their ideas and users to try them out




- Different kinds of prototyping
  - Low-fidelity
  - Medium-fidelity
  - High-fidelity

# What is a prototype?

- A prototype is an **approximation** of the product on one or more **dimensions of interest**
- Is one manifestation of a design that **allows stakeholders to interact with it**
- In interaction design a prototype can take the form of (among other things):
  - a series of screen sketches
  - a storyboard, i.e., a cartoon-like series of scenes
  - a Powerpoint slide show
  - a video simulating the use of a system
  - a cardboard mock-up
  - a piece of software with limited functionality

# Why a prototype?

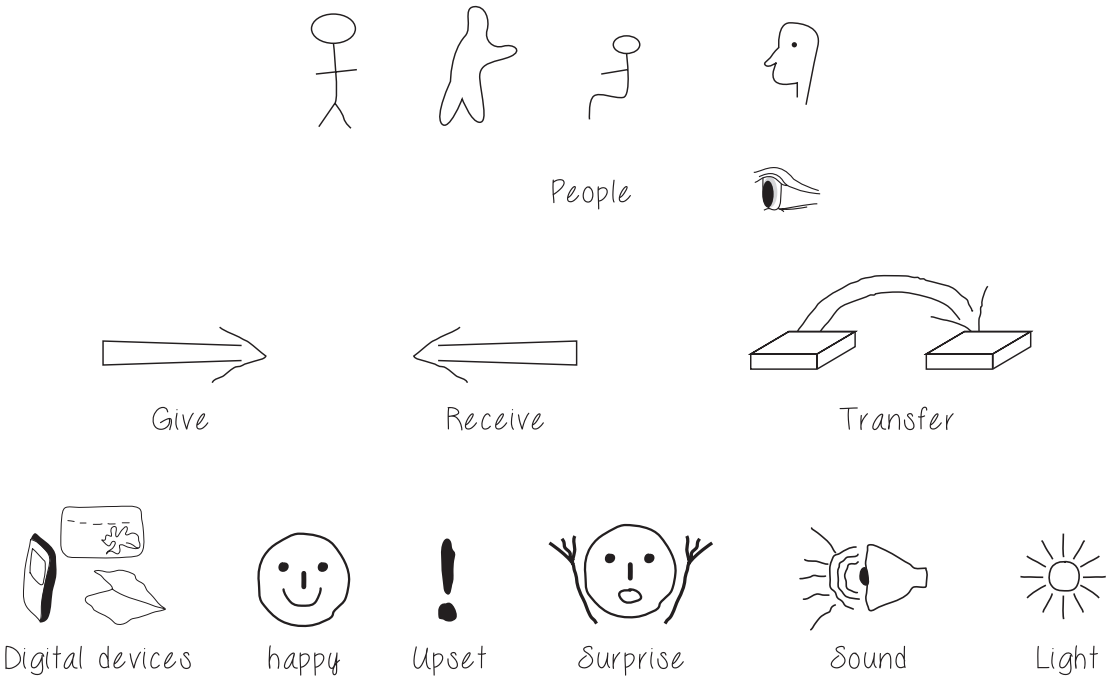
- Evaluation and feedback are central to interaction design
- Stakeholders can see, hold, interact with a prototype more easily than a document or a drawing
- Team members can communicate effectively
- Ideas can be tested out: **fail early** to avoid disasters later
- Prototyping encourages reflection: an important aspect of design
- **Prototypes answer questions, and support designers in choosing between alternatives**

# Low-fidelity prototyping

- May use a medium which is unlike the final medium, e.g., paper, cardboard
- Is quick, cheap and can be easily changed
- Examples:
  - **Sketches** (of screens, task sequences, etc.)
  - **Storyboards**
  - **"Wizard-of-Oz"**

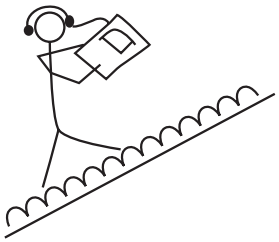
# Sketching

- Low-fidelity prototyping often relies on hand-drawn sketches
- Don't be inhibited about drawing ability. Practice simple symbols
  - Sketching is not about drawing, it is about design



# Storyboards

- It is a series of sketches showing how a user might progress through a task using the product, plus word descriptions
- Often used with scenarios, bringing more detail, and a chance to role play



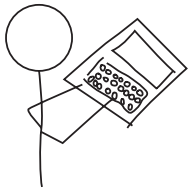
Christina walks up hill; the product gives her information about the site



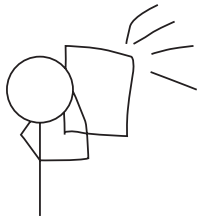
Christina adjusts the preferences to find information about the pottery trade in Ancient Greece



Christina scrambles to the highest point



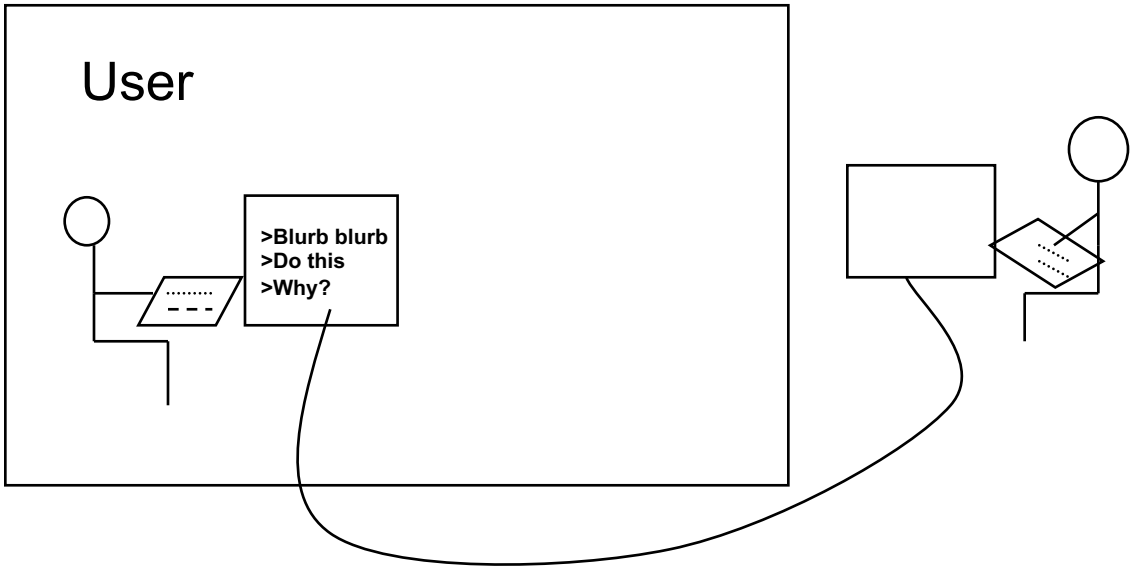
Christina stores information about the pottery trader's way of life in Ancient Greece



Christina takes a photograph of the location of the pottery market

# "Wizard-of-Oz" prototyping

- The user thinks they are interacting with a computer, but **a human is simulating** the responses of the system
- Usually done early in design to understand users' expectations



# "Wizard-of-Oz" prototyping



- **Some aspects of interface are implemented** (but not all)
  - The users' operations require processing that is actually done by a human **not visible to the user**
  - The "wizard" intervention needs to be **believable**

# High-fidelity prototyping

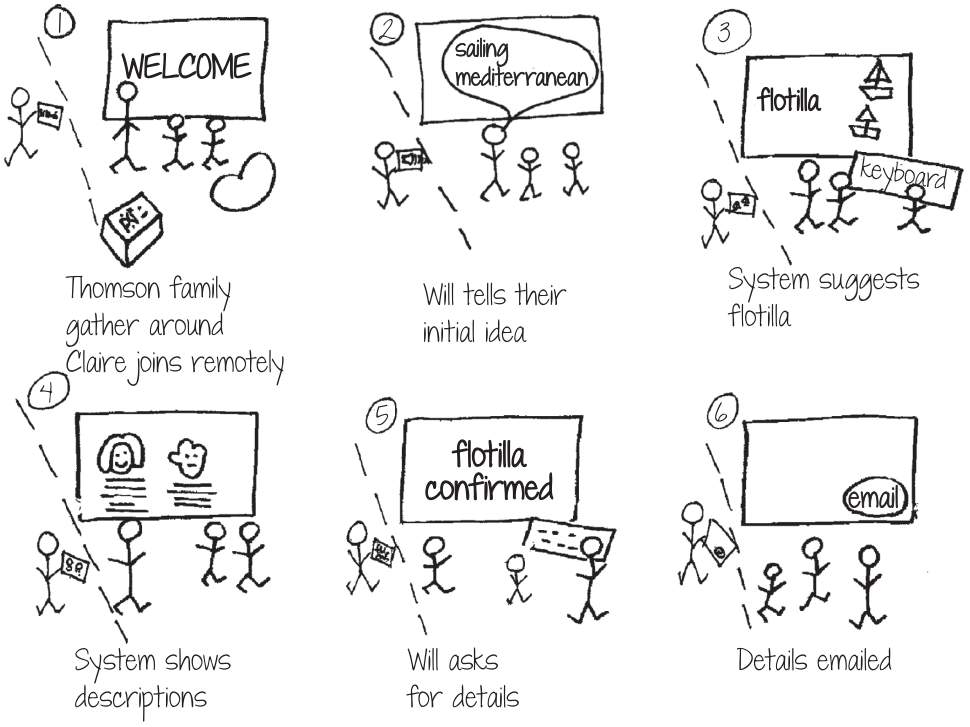
- Prototype looks more like the final system than a low-fidelity version
- Uses materials that you would expect to be in the final product
- High-fidelity prototypes can be developed by **integrating** existing hardware and software components
  - Which you have tested in previous low- or medium-fidelity prototypes

# Compromises in prototyping

- **Prototyping involve compromises**: the intention is to produce something quickly to test an aspect of the product
- Two common types of compromise
  - **Horizontal**: provide a wide range of functions, but with little detail
  - **Vertical**: provide a lot of detail for only a few functions
- **Compromises in prototypes must not be ignored** (e.g., be aware of them in the design and evaluation)
- The kind of questions that any one prototype can answer is **limited**, and the prototype must be built with the key issues in mind

# Generating low fidelity prototypes

- Prototypes are generated from the output of the requirements activity
- Now we see how to generate a **storyboard prototype** from a scenario

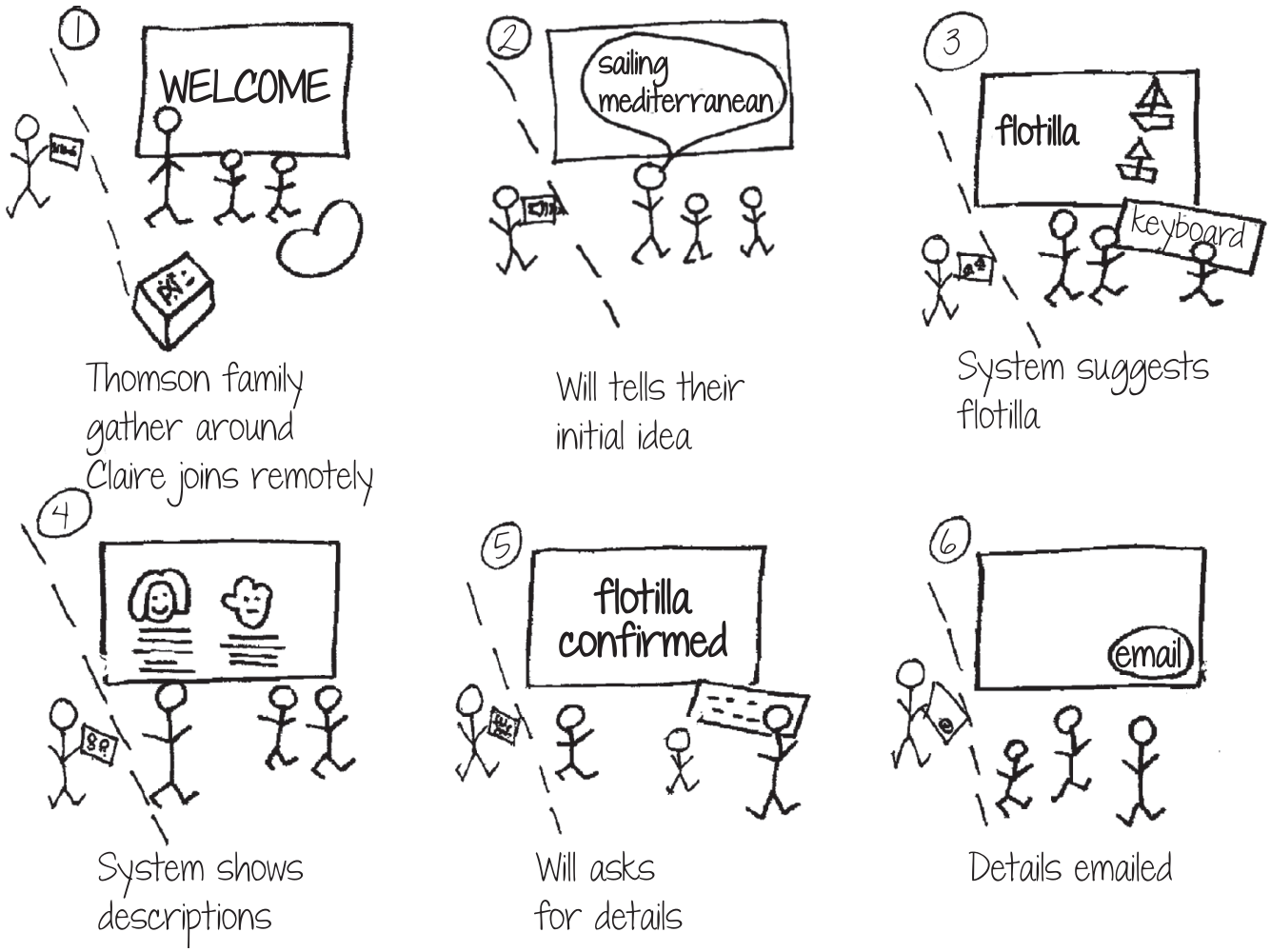


# Generating storyboards from scenarios

- A **storyboard** is a low-fi prototype that visually represents a sequence of actions or events that the user and the product go through to achieve a goal
- A **scenario** is one story about how a product may be used to achieve that goal
- To generate a storyboard from a scenario
  1. **Break down the scenario into a series of steps**
  2. **Create one scene for each step**
- A storyboard
  - can be used to get feedback from users and colleagues
  - prompts the design team to consider the scenario and the product's use in more detail

# Example of storyboard generated from a scenario

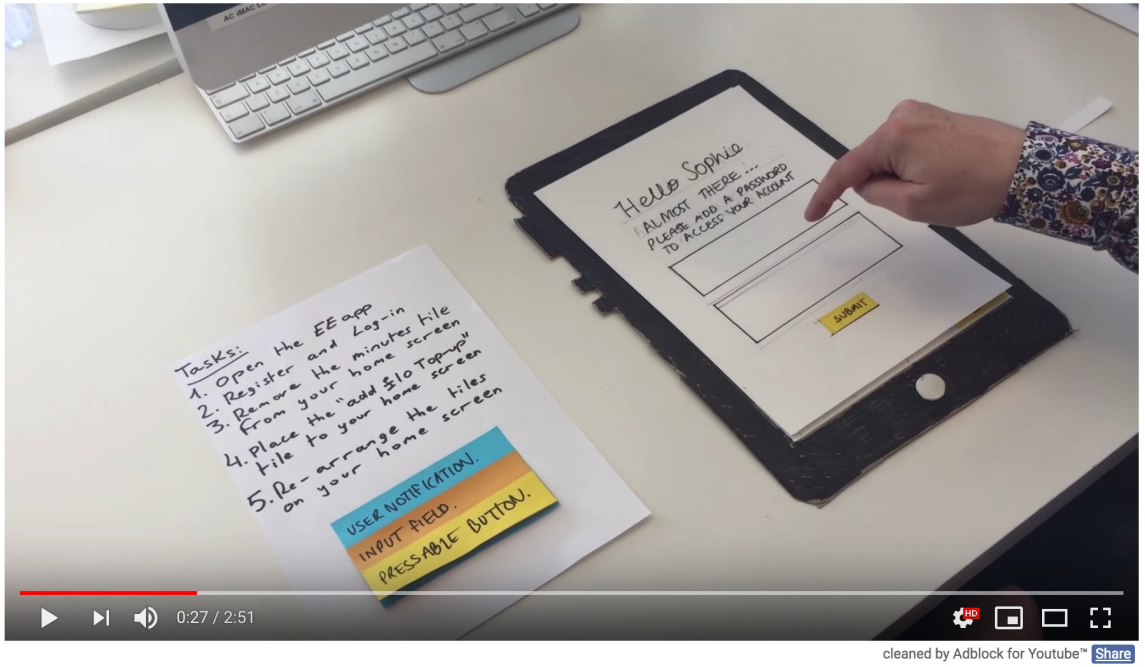
- The scenario has been broken down into 6 steps



# Example of paper prototype in action

- A set of card-based prototypes that cover a scenario from beginning to end may be the basis of a more detailed prototype, such as a paper prototype of an interface

Video of a paper prototype in action used in an **evaluation session**

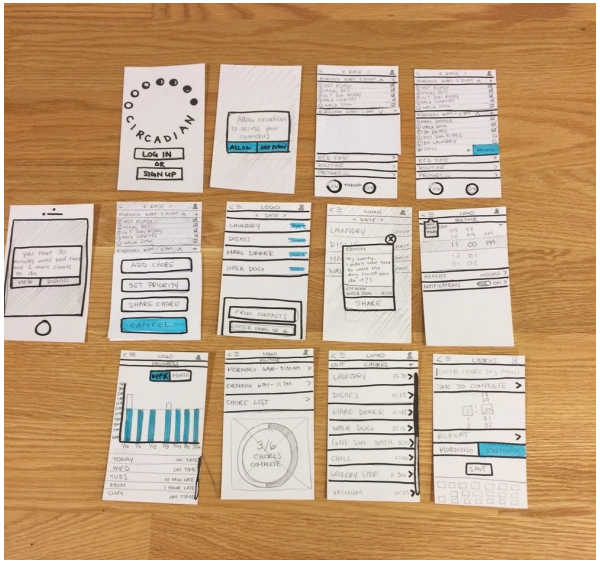



Low fidelity prototype testing of the EE app

https://www.youtube.com/watch?v=yafaGNFu8Eg

# Paper prototypes

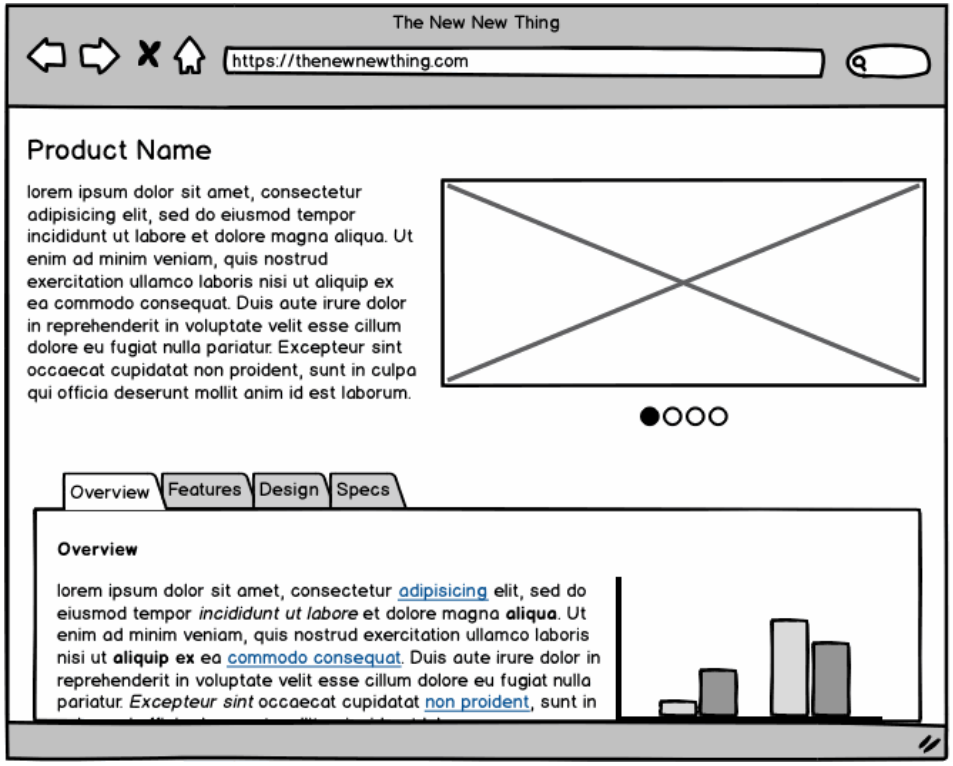
- **Advantages**
  - Quick to build/refine, thus enabling rapid design interactions
  - Requires minimal resources and materials




- **Disadvantages**
  - It is paper not a real app/webpage

# Wireframes

- A wireframe is a static, low-fidelity representation of your product
- In the world of web and mobile design, is a basic outline of your website or app – **the skeleton**



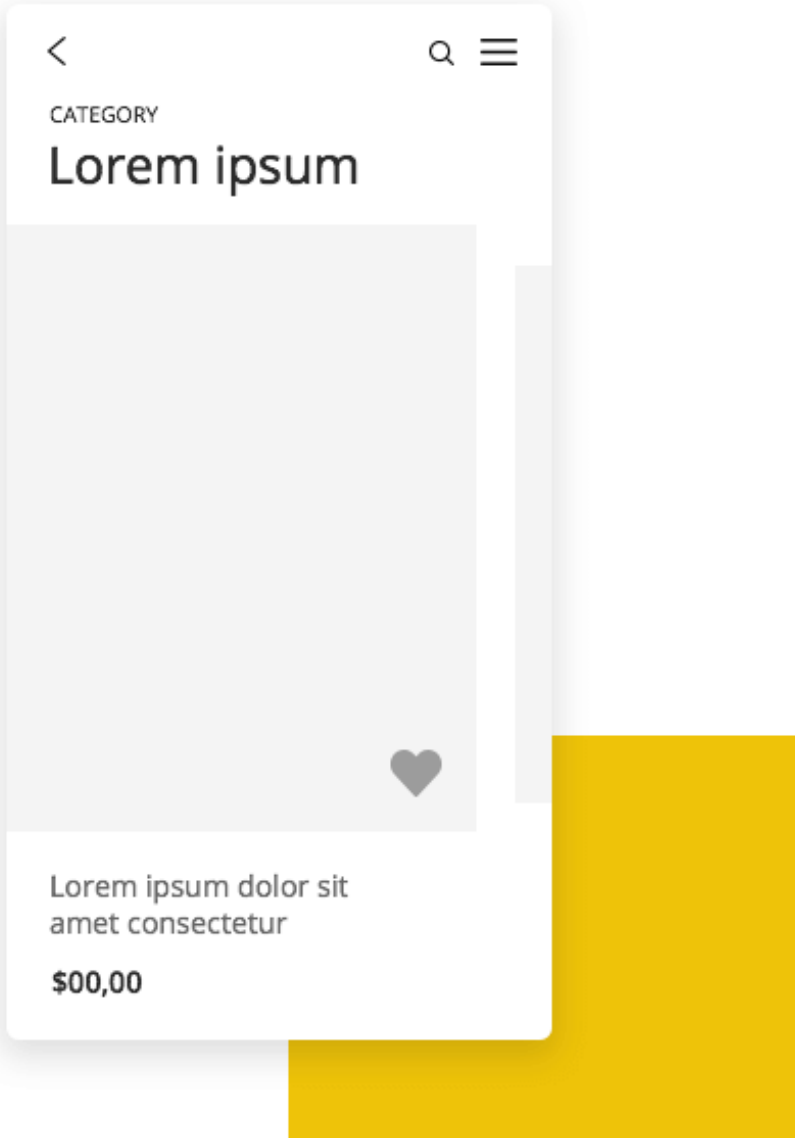

- With wireframes, you dedicate time **solely** to answering crucial layout, structure, and organization questions before the team iterates on visual details

# Benefits of wireframes

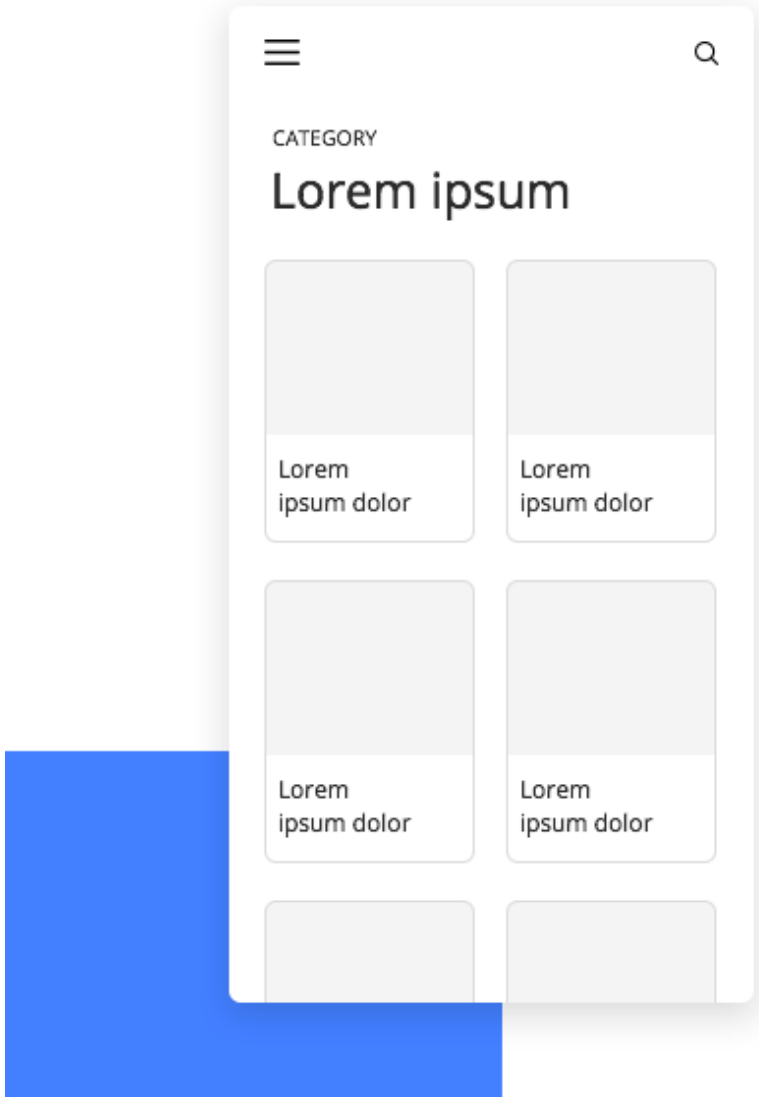
WIREFRAMING

## Benefits

- Wireframes are quick, cheap and easy to create
- Focus on the big picture
- Easy for stakeholders to visualize and digest



# Limitations of wireframes



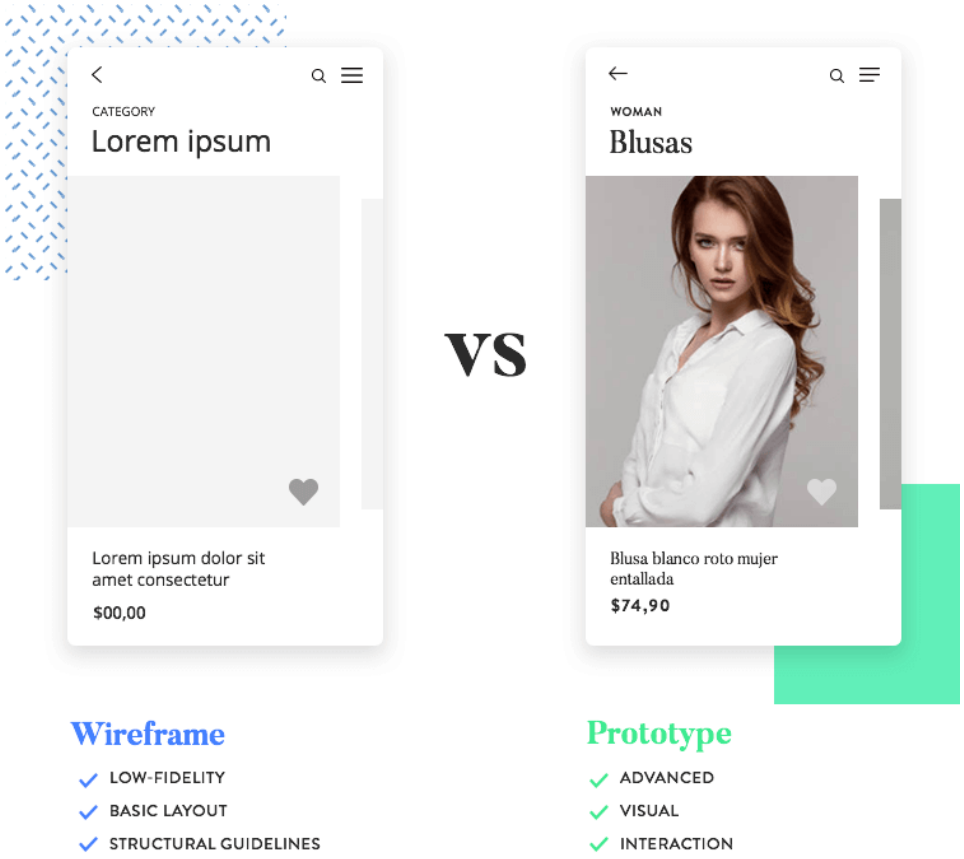

WIREFRAMING

## Limitations

- Only Lorem ipsum. Not real content.
- Not used to test or validate.
- Without branding.

# Wireframes vs prototypes

- Wireframes are low-fidelity, basic layout and structure
- Prototypes are an advanced wireframe with **more visual detail** and interaction




- Wireframes just help you to focus the placement of content for your prototype
- Wireframes are simply a means to achieve the prototype

# Wireframing tools

- Sketches via pen and paper

- Digital wireframes using software tools
  - We will use Axure RP
    - Is a tool that allows to create both wireframes and interactive prototypes
    - https://www.axure.com
    - Free for students

Lesson for the course “Human-Computer Interaction”

# Axure RP (Part 1)

Prof. Luca Turchet

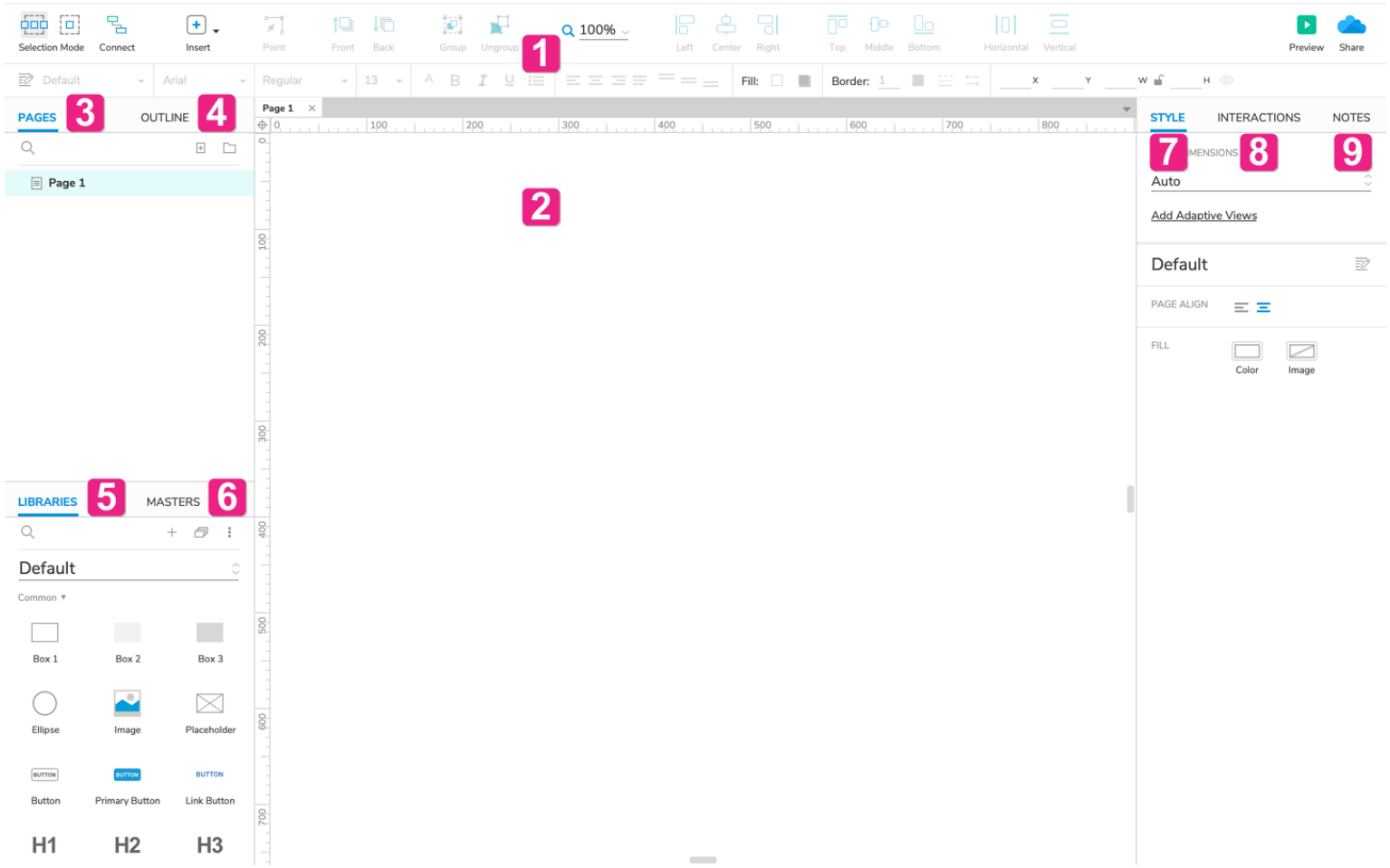
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# Introduction to Axure

- Features:
  - **No programming required**: create simple or highly functional prototypes with conditional logic, dynamic content, animations without writing a single line of code
    - Just drag the components or interactions that you need from the palettes to the work area
  - **Instant simulations**: see your application's prototype in action
  - **Exportable to HTML**: export to HTML effortlessly, so that your clients or users can see it work online and give you their opinion
- Getting started
  - https://docs.axure.com/tutorials/basics/getting-started/

# Axure: the environment

- https://docs.axure.com/axure-rp/general/environment/



# Axure: getting started

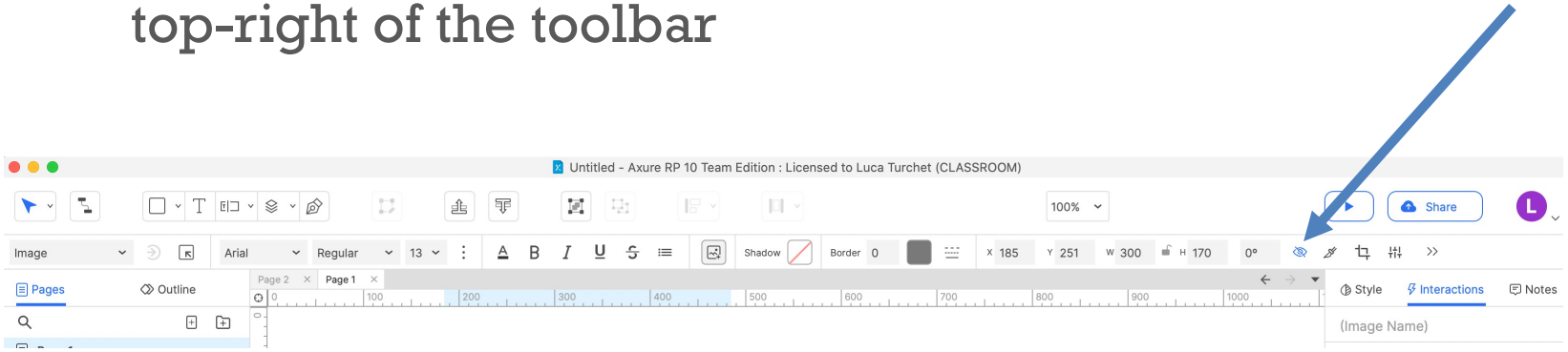
- **Pages**
  - Axure prototypes are organized into pages
  - https://docs.axure.com/axure-rp/pages/managing-pages/
- **Canvas**
  - It is where you draw your diagrams
  - You can set the canvas for a target device (e.g., iPhone X)
- **Widgets**
  - You draw diagrams using widgets
- **Interactions**
  - You add interactivity to the widgets: e.g., on-click go to another page
- **View** your prototype on a browser or a mobile app (download it)
- **Share** using for free the Axure cloud service
- **Documentation/Notes**

# Axure: Create a button that goes to another page

1. create two pages in the Page pane
2. on both pages create a button using the widget button (drag and drop from the Libraries Pane)
3. name each button differently
4. select the button on page 1 and select "New Interaction" in the Interactions pane
5. select ”Click or Tap" and  select "Open Link", then select Page 2 from the drop down
6. analogously for the button on page 2
7. Preview the page to test out the interaction

# Axure: Display an image when you mouse over a button and hide it when you mouse off the button

1. add a widget for a button and a widget for an image
2. select the image widget and select the "hidden" icon in the top-right of the toolbar




3. select the button and select "New Interaction" in the Interactions pane
4. select the "Mouse Enter" event and then the "Show/Hide" action

# Axure: Display an image when you mouse over a button and hide it when you mouse off the button

5. in the dropdown select the Image widget and select the "Show" field, then click Ok
6. select the button and select "New Interaction" in the Interactions pane
7. select the "Mouse Exit" event and then the "Show/Hide" action
8. in the dropdown select the Image widget and select the "Hide" field, then click Ok
9. Preview the page to test out the interaction

# Axure: change the widget properties using the Style effect: make the button turn blue when you mouse over it

1. Right click the button and select ”Style Effects”
2. Select MouseOver
3. In the panel, select "Fill color" and then pick the color red
4. Preview the page to test out the interaction

# Axure: Share with your team your prototype: publish it on the Axure Cloud

1. Click the share button
2. Optionally, give to your project a name and a password
3. Share the link with your team (use the top right button)

# Homework

- Hear some interesting words from UX designers at Google about paper prototyping:
  - https://www.youtube.com/watch?v=JMjozqJS44M
- **Get acquainted with Axure**
- From scenario to protype:
  1. Invent a simple scenario
  2. Generate a storyboard from it
  3. Create a wireframe for a simple mobile app
  4. Crete a paper prototype
  5. Quickly test it with a user
  6. Implement it in Axure

# Bibliography


- Chapters 3 and 12 of the book "Interaction Design: beyond human-computer interaction" - 5th Edition, by Sharp, Rogers, and Preece. Wiley, 2019
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